

7.6 Nitrogen Compounds (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.6 Nitrogen Compounds (A Level Only)
Difficulty	Easy

Time allowed: 20

Score: /13

Percentage: /100

Question 1a

This question is about the synthesis and nature of amines.

State the reagents and conditions required for the conversion of 1-bromobutane into butylamine.

[3 marks]

Question 1b

Explain why amines can act as bases.

[1 mark]

Question 1c

Write an equation to show how ethylamine acts as a base when reacted with hydrochloric acid and give the name of the product formed in this reaction.

[2 marks]

Question 2a

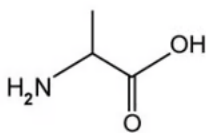
This question is about amino acids.

State the general formula for an amino acid.

[1 mark]

Question 2b

The amino acid alanine is shown in Fig. 2.1



Alanine

Fig. 2.1

Draw the **two** structures of alanine in acidic and basic conditions.

acidic conditions	basic conditions
-------------------	------------------

[2 marks]

Question 2c

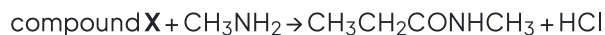
Electrophoresis is an analytical technique which separates ions by placing them in an electrical field.

Describe briefly how a sample of an amino acid can be separated using this technique.

[1 mark]

Question 3a

Amides can be formed from amines. An example is shown in the following reaction scheme where compound **X** reacts with methylamine to form *N*-methyl propanamide and hydrogen chloride.



State the structural formula of compound **X**.

[1 mark]

Question 3b

N-methyl propanamide can undergo reduction forming compound **Y** and water. State the reagent required for this reaction.

Reagent

[1 mark]

Question 3c

A different amide, ethanamide was heated with sodium hydroxide the gas ammonia is formed along with one other product.

i)

Write the equation for this reaction.

[1]

ii)

Explain how this reaction could be used as a test for an amide.

[2]

[1 mark]

Model Answers: Easy

1a

a) The reagents and conditions required for the conversion of 1-bromobutane into butylamine

- Excess ammonia; [1 mark]
- Ethanolic/ ethanol as a solvent; [1 mark]
- Hot and under pressure; [1 mark]

[Total: 3 marks]

- Amines can be prepared by:
 - The reaction of halogenoalkanes with ammonia
 - The reduction of nitriles
- Make sure that you know the reagents and conditions required for both of these reactions

1b

b) Amines can act as bases as:

- They can act as proton acceptors by forming a dative covalent bond with an H^+ ion
OR
They can donate their lone pair of electrons to an H^+ ion forming a dative covalent bond; [1 mark]

[Total: 1 mark]

- Bases can be defined as proton acceptors or electron donors
- An amine can accept a proton (H^+ ion) by donating its lone pair of electrons - so amines are bases

1c

c) An equation to show how propylamine can act as a base if reacted with hydrochloric

acid is:

- $\text{CH}_3\text{CH}_2\text{NH}_2 + \text{HCl} \rightarrow \text{CH}_3\text{CH}_2\text{NH}_3^+ + \text{Cl}^-$; [1 mark]
- Ethylammonium chloride; [1 mark]

[Total: 2 marks]

- The product you write must show the correct charges on the correct atoms
- The nitrogen atom in propylamine can accept a proton (H^+ ion)
- It can therefore act as a Brønsted-Lowry base by donating its lone pair of electrons to a proton and forming a dative bond
- Amines react with HCl to form amine salts, just as a standard acid-base reaction produces a salt

2a

a) The general formula for an amino acid is:

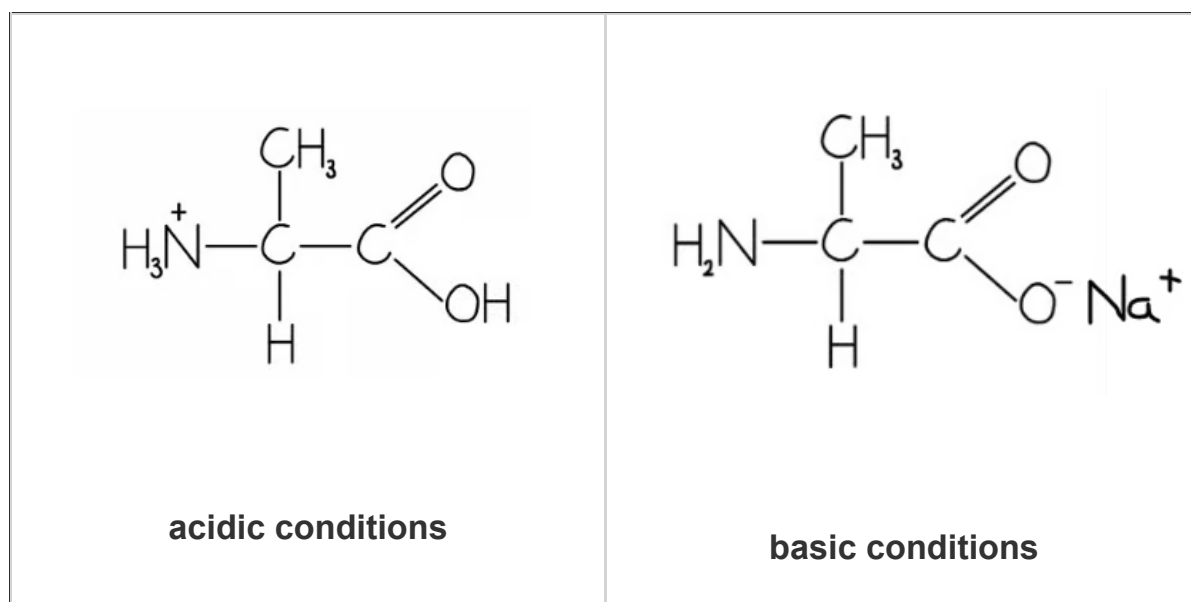
- $\text{RCH}(\text{NH}_2)\text{COOH}$; [1 mark]

[Total: 1 mark]

- You are expected to know the general formula for an α -amino acid

2b

b) The structures of alanine in acidic and basic conditions are:



- Each correct structure scores [1 mark]

[Total: 2 marks]

- Amino acids have an acidic COOH group
- The -COOH group has a tendency to lose a proton (act as an acid)
 - So in basic conditions: $\text{-COOH} \rightleftharpoons \text{-COO}^- + \text{H}^+$
 - This can react with the positive ion from the alkali to form a salt
- Amino acids have a basic NH₂ functional group
- The amine group has a tendency to accept a proton (act as a base)
 - So in acidic conditions: $\text{NH}_2 + \text{H}^+ \rightleftharpoons \text{NH}_3^+$

2c

c) A sample of an amino acid can be separated using this technique by:

- A sample of amino acids is placed between two oppositely charged electrodes; [1 mark]
- The positively charged ions will move towards the negative electrode
AND
The negatively charged ions will move towards the positive electrode; [1 mark]
- Larger ions move more slowly
OR
Smaller ions move more quickly; [1 mark]
- Highly charged ions move more quickly
OR
Ions with a low charge move slowly; [1 mark]

[Total: 4 marks]

- The question asks you to briefly explain how amino acids can be separated in this technique, not identified
- To identify them the electropherogram is used
- This is the series of bands which are observed on the paper or gel after electrophoresis has occurred
 - Each band in the electropherogram corresponds to a particular species

3a

a) The structural formula of compound **X** is:

- $\text{CH}_3\text{CH}_2\text{COCl}$; [1 mark]

[Total: 1 mark]

- Amides can be formed by the reaction of an amine with an acyl chloride
- In this case a primary amine, CH_3NH_2 , has been reacted with propanoyl chloride to form the substituted amide and hydrochloric acid

3b

b) The reagent required and name for compound **Y** is:

- LiAlH_4 ; [1 mark]

[Total: 1 mark]

- *N*-methyl propanamide is a substituted amide, so the amine formed via the reduction is a secondary amine and water
- Make sure you know reducing agents
 - LiAlH_4 is a strong enough reducing agent for this reaction whereas NaBH_4 is not

3c

c)

i) The equation for the reaction of *N*-methyl propanamide with sodium hydroxide is:

- $\text{CH}_3\text{CONH}_2 + \text{NaOH} \rightarrow \text{CH}_3\text{CO}_2\text{Na} + \text{NH}_3$; [1 mark]

ii) This reaction can be used as a test for amides by:

- Test for ammonia gas; [1 mark]
- Damp red litmus paper turns blue; [1 mark]

[Total: 3 marks]

- If ethanamide is heated with sodium hydroxide solution, ammonia gas is given off and you are left with a solution containing sodium ethanoate
- As the reaction forms ammonia gas you can test for this using damp red litmus